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ABSTRACT

The aim of the present study was to investigate how quickly and efficiently distracted drivers react to four different visual warning displays intended for a Forward Collision Warning system for vehicles. The displays are the Collision Warning Head Up Display (CWHUD), the Steering Wheel Warning Display, The High Head Down Display (HHDD), and the Cluster display. The study was performed in a simulator.

Warnings were randomly presented to drivers when they were visually distracted away from the road by a traffic sign discrimination task. Brake reaction times to the different warnings were recorded. From the results of performance tests and preference ratings, the CWHUD, provided the best performance in terms of 200 ms lower reaction time and low amount of missed warnings. The average brake reaction times were significantly faster for the CWHUD warnings. There were no significant differences in reaction times among the other three warnings. One CWHUD warning was missed during the experiment, whereas 17 High Head Down Display, HHDD, warnings were missed. The CWHUD was rated first among the user preferences. The second, third and fourth choices were HHDD, Cluster Display, and Steering Wheel Warning Displays. It is concluded from the results of this study that the CWHUD was the best of the four FCW visual displays tested in the study.

The high performance and high subjective driver preference is contributed to the location color and brightness of the CWHUD providing the driver with an intuitive understanding of a forward braking vehicle.

INTRODUCTION

The present study investigates the performance of different Forward Collision Warning (FCW) displays in a driving simulator. FCW systems provide alerts to assist drivers in avoiding or mitigating rear-end crashes. A warning must provide a fast and appropriate response from drivers in critical close to impact situations. The design of the warning displays for FCW is very

important for safety because a poorly designed display may reduce its effectiveness.

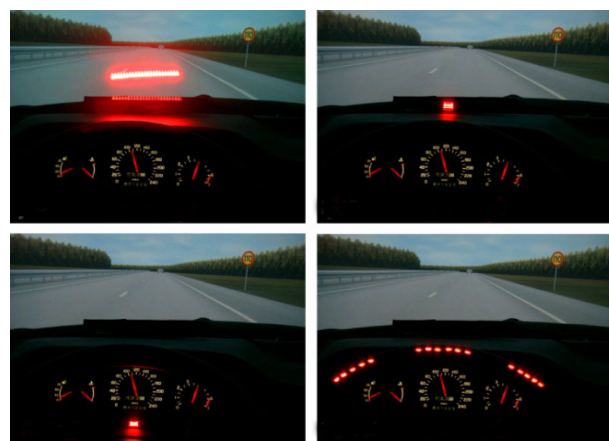


Figure 1 *Warning displays in study*

It has been demonstrated that visual warnings are an essential component of the single stage FCW¹ however there is much uncertainty about how and where this warning should appear. Conspicuity is an important issue in the design of warning displays. Cole and Hughes² discuss issues relating to visibility or target conspicuity while driving. Conspicuity is "...the property that leads to a target object having a high probability of being seen within a very short time"². They summarize the following factors that determine conspicuity: eccentricity from line of sight, contrast between target and background, complexity of background, color and the boldness of information on the target. For maximum conspicuity, a FCW visual warning display should be near to a driver's line of sight, have high contrast against a uniform background, and be bright and colorful. Unfortunately it is not that simple, particularly for the line of sight issue.

Guidelines recommend that visual warnings be placed within 15 degrees of the driver's normal sight line. The vast majority of driver's visual attention is on the forward road scene (roughly 85-90%³). The visual warning should be easily visible when the driver is looking

forward. More than 75% of the drivers are non-attentive to the road-way before a frontal collision according to the 100 cars naturalistic driving study⁷. If they were looking at the forward road scene, they would be able to see and avoid a hazardous situation. Therefore an FCW system should be seen as an intelligent "Anti-distraction system" and must provide the property of directing the sight of a distracted driver to the forward road scene.

Where do drivers look when they are distracted (i.e., not looking at the road)? They could be checking their mirrors, looking ahead but not attending, interacting with an information system (e.g., radio), or perhaps talking with a passenger. According to field research by Lansdown³ on drivers in the U.K., the mean percentage of time spent looking to the different regions of the visual scene were: 4.6 % rear-view mirror, 2.8 % right region, 1.3 % left region and 3.9 % instrument panel & centre stack. It is important that when drivers are looking in these four areas, and not looking at the road, the FCW display must also be visible. Meanwhile, the display should still be central to the forward view because the warning must direct the driver's attention to the hazard. A disadvantage of highly conspicuous visual displays is that they can capture attention rather than direct the eye to a hazard ahead.

WARNING DISPLAY CHARACTERISTICS

Two new warning display concepts, the Collision Warning Head Up Display, CWHUD, (figure 2) and the Steering Wheel Display (figure 5), with the aim to provide visibility, attention and intuitiveness.

A selection of two different visual warning displays was made from literature and from market available systems. These were the High Head Down Display, HHDD (figure 3) and the Cluster Display (figure 4).

Table 1 to 4 lists the characteristics of the warning displays.

A flashing red warning was selected to allow for a good peripheral recognition of the activation.

Each warning lasted 1.2 second and flashed at a rate of 4 times per second. The on flash lasted 0.15 seconds with 0.1 seconds between flashes. The table below describes how the displays differed. Figure 1 shows pictures of how the four different warnings appeared to the driver.

COLLISION WARNING HEAD UP DISPLAY, CWHUD



Figure 2 *Collision Warning Head Up Display*

Position:	Top of instrument panel reflected in lower part of windscreen
Appearance:	Red light
Intensity measured from eye position:	1000 mcd
Size:	14.5 x 0,5 cm
Light source:	23 round clear red high intensity LED's of 1000mcd each. High intensity is required since only roughly 10% of the visual intensity is reaching the eye through the reflection in windshield.

Table 1 *Collision Warning Head Up Display characteristics*

HIGH HEAD DOWN DISPLAY, HHDD



Figure 3 High Head Down Display

CLUSTER DISPLAY



Figure 4 Cluster Display

Position:	Direct view display on top of instrument panel.
Appearance:	Telltale & warning in text (<i>CAMP symbol</i>)
Intensity measured from eye position:	80 + 80 mcd covered with icon frame
Size:	2.0 x 1.0 cm
Light source:	2 red back-lighters composed of 16 LEDs

Table 2 High Head Down Display characteristics

Position:	Direct view, In instrument cluster
Appearance:	Telltale & warning in text (<i>CAMP symbol</i>)
Intensity measured from eye position:	80 mcd - covered with icon frame
Size:	1.5 x 0.7 cm
Light source:	1 red back-lighter composed of 8 LEDs

Table 3 Cluster display characteristics

STEERING WHEEL DISPLAY



Figure 5 *Steering Wheel Display*

Position:	Direct view, Top centre of steering wheel rim
Appearance:	Lights
Intensity measured from eye position:	80 mcd
Size:	26 cm x 0.2 cm
Light source:	16 square LEDs grouped as 5, 6 and 5.

Table 4 *Steering Wheel Display characteristics*

METHOD

Sixteen experienced drivers aged 24 to 60 years (mean age = 37 years) participated in the study. Nine were female and seven were male.

The study was run in a stationary base driving simulator to be able to measure the performance in a controlled environment. Participants were given a five-minute practice drive to become familiar with the driving simulator. The test drive lasted for approximately 15 minutes on a motorway. Participants were instructed to drive as they normally would, but to maintain the speed limit (110 km/h). A forward vehicle was cruising at fixed distance to the ego vehicle approximating about 1.5 second's headway. There was other traffic, but nothing that interfered with the driver's speed.

During the test drive, participants were presented with random targets in four different locations. The targets are provided in the following pictures taken from the driver position. Figure 6a shows a target in the Centre stack display. Figure 6b shows the target in the far left of the projected road scene approximating the left mirror position. Figure 6c shows the image in the far right approximating the right mirror position and finally figure 6d shows the image in high centre approximating the inner rear view mirror position. The targets were five different road signs that appeared in one of the locations every four seconds on average. Drivers were asked to respond, with a button press, to the appearance of one of five target options (the deer sign). Target detection was rewarded with points. Approximately 12 responses per target position were recorded amongst about 225 total signals.

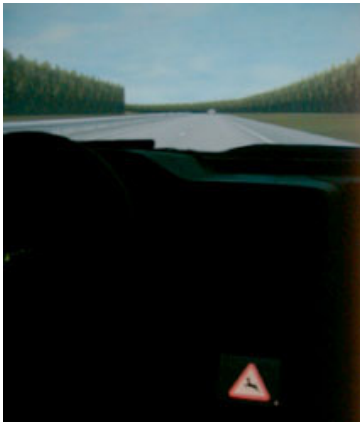


Figure 6a *Centre stack position*

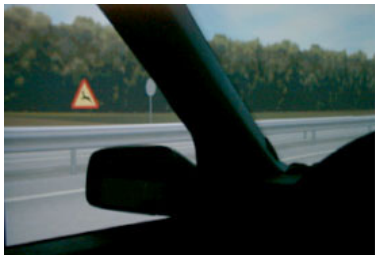


Figure 6b *Left mirror position*



Figure 6c *Right Mirror position*



Figure 6d *Rear view mirror position*

The number and frequency of targets were selected to create a realistic amount of visual distraction. FCW warning events were distributed where the visual warning display would light up and the drivers were instructed to press the brake as fast as possible. A warning could only occur 0.5 seconds after a target sign was displayed so the driver would be looking at the sign when the warning light appeared. The drivers were warned every 30 seconds on average with a minimum spacing of at least 15 seconds between signals. The drivers' brake reaction times to the FCW warnings were timed. Approximately 8 responses per warning type were recorded.

All of the reaction time data for both the warnings and targets were recorded in a computer log file. The reaction time data for each participant were summarized and transcribed into Statistica where they could be further analyzed.

RESULTS

Repeated measure ANOVAs were conducted on the reaction times to the four warning lights. There were significant differences between the warning lights for the mean [$F(3, 45) = 10.8, p < 0.0001$], median [$F(3, 45) = 11.4, p < 0.0001$] and maximum [$F(3, 45) = 4.28, p < 0.01$] reaction times. Post-hoc comparisons using Tukey's HSD indicated, for these measures, that the reaction times were significantly faster for the CWHUD than the other three warning displays, figure 7. This difference was nearly 200 ms for the mean brake reaction times.

The standardized effect between the CWHUD and Cluster warnings was 0.87, which constitutes a large difference. No significant differences were found among the other warning lights and there were no gender differences in reaction time.

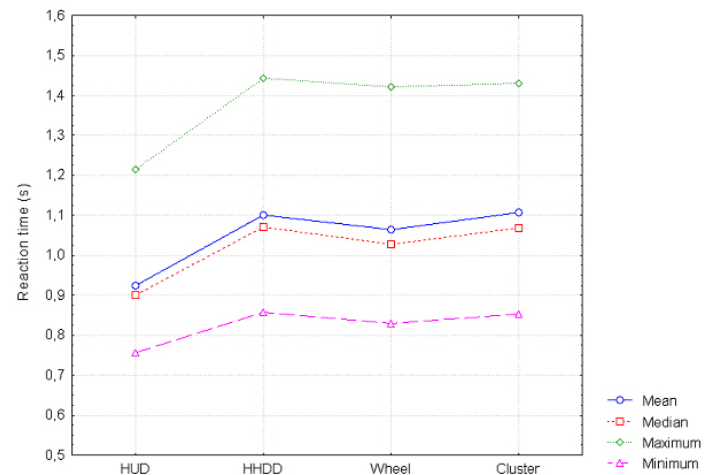


Figure 7 *Reaction time to warning*

Regarding the number of participants who missed a warning, only one driver missed a HUD warning, 6 drivers missed a Steering Wheel warning, 5 missed a HHDD warning and 11 missed Cluster warnings. Some people missed more than one warning. The number of missed warnings per display distributed as an average per driver is shown in figure 8. The total number of missed warnings was as follows: 1 CWHUD, 6 HHDD, 12 SW and 17 Cluster warnings. Paired comparisons with the Wilcoxon matched pairs test indicated that this difference was statistically significant between the CWHUD and other warnings ($\alpha = 0.05$).

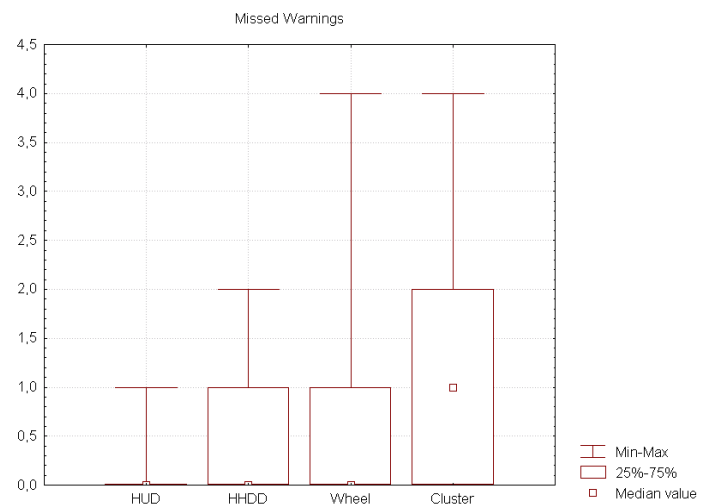


Figure 8 *Missed warnings per warning display*

Repeated measure ANOVAs were conducted on the average reaction times to the four target positions. There were significant differences between the four locations [$F(3, 39) = 27.0, p < 0.0001$; note two

participants were excluded from this analysis because they missed all the targets that appeared on the centre stack monitor]. Post-hoc comparisons using Tukey's HSD indicated that the reaction times were significantly faster for the middle screen targets than the other 3 locations. Also the right screen targets had significantly faster responses than the left screen position.

The participants were asked to rank the four warning lights according to their preference at the end of their drive, see figure 9. Most preferred the CWHUD warning light overall followed closely by the Steering wheel. The HHDD was third choice and Cluster warning was the least preferred option.

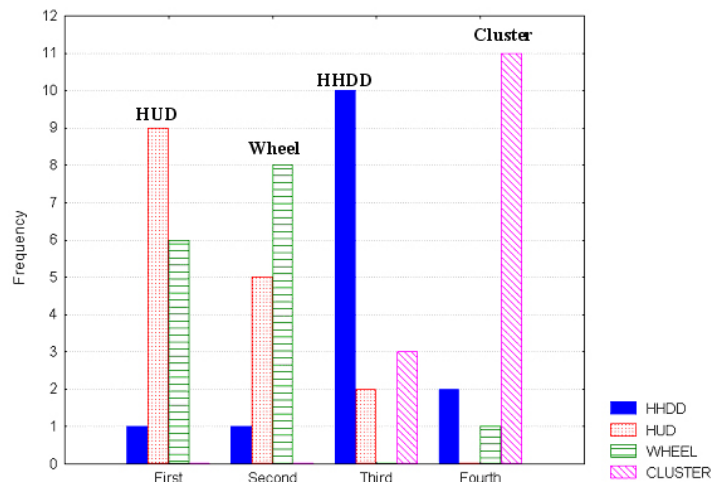


Figure 9 *Ranking of visual warning display*

SOURCES OF ERROR

Before concluding with some recommendations, some limitations of this study should be discussed. One limitation was that the warnings were given during 'normal' driving rather than collision scenarios (e.g., sudden braking vehicle ahead). This may have surprised drivers because they had no indication of when the warning would appear. An opposing problem was the test drivers were expecting the warning throughout the drive, so they may have responded faster than they would have in a real collision-warning situation. On the balance of these factors, it is believed that this study obtained a reasonably accurate measure of the speed to which distracted drivers react to collision warning displays since the results are comparable to the CAMP study results¹ of 1.18 seconds 85 percentile reaction time for a combined HHDD and audible warning. The aim of the study was to make an interface comparison and not to define the reaction time. For this purpose the method is believed to be fully valid.

The simulator setup with lower environmental brightness than a real world scenario may improve the performance. However the relationship in brake reaction time between warning displays may still be similar.

The driver foot position before warning was not recorded in the study. However since all drivers experienced the same warnings the distribution of error for different drivers foot position should be evenly spread over the warning displays.

Another limitation of this study was that, because the four warnings were unique, no conclusions regarding the ideal design for a FCW can be made. For example, little can be said about the general effectiveness of LED's versus telltales, or high versus low warnings. The study can only say what is best among a limited selection of visual warning types rather than what is the best possible visual warning for FCW.

CONCLUSION

From the results of the performance tests and preference ratings, the CWHUD display was in the study found to be the best for the FCW. The average brake reaction times were significantly faster for the CWHUD warnings (approximately 200 ms). The same result was found for the median, minimum and maximum reaction times. There were no significant differences in reaction times among the other three warnings. The CWHUD also performed best related to low amount of missed warnings. Only one CWHUD warning was missed during the experiment, whereas 17 of the total number of instrument Cluster warnings were missed by the drivers. Also, the CWHUD rated first among the user preferences. The second, third and fourth choices were Steering wheel, HHDD, and Cluster.

The cluster position had the worst performance and preferences ratings. Also, it was the only warning display that was not within 15 degrees of the driver's normal site line which is a NHTSA recommendation⁴.

The explanation for the performance of the CWHUD is the greater size and brightness combined with the similarity and the position of a red brake light that together provide driver with an intuitive understanding of the situation.

Not considered in this study was the problem of driver annoyance. Frequent false alarms or nuisance warnings may cause the drivers to become irritated and this risk of irritation increases when the warnings are intrusive. Further studies preferably in realistic on road driving conditions on should be done to ensure that an efficient FCW visual display is not too annoying under normal realistic operating conditions which include false and nuisance alarms. A system utilizing an intrusive type of Collision Warning Display is expected to demand less nuisance warnings and a very low amount of false warnings than a low intrusive Collision Warning system.

The use of complementary warning modalities can not be ruled out as efficient means of warning. I.e. less intrusive visual warning signals like the HHDD, Steering Wheel Display and Cluster Display used together with

an audible warning can not be ruled out as an efficient collision warning.

Low sunlight may in real life reduce the visibility of the warning as well as if the CWHUD warning is outside the peripheral vision of the driver. It is therefore recommended to combine the CWHUD with a complementary warning modality.

Internal studies on the CWHUD have been performed in more realistic FCW scenarios both in simulator and on road. These studies show that the proposed CWHUD visual warning device is efficient and intuitive in collision scenarios.

To conclude the results shows that the Collision Warning Head Up Display, CWHUD, provided the best performance in a distracted driver situation based on a low amount of missed warnings, low reaction time and high driver preference out of the four FCW visual displays tested.

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